ME228 — Applied Data Science and Machine Learning — *Final Project*

Data-Driven Fatigue Strength Prediction and Inverse Alloy Design

Mixture-of-Experts Regression with Empirical Reliability & Cost-Constrained Pareto

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1 Executive Summary

Fatigue causes 80–90 % of mechanical failures [1], yet fatigue tests are destructive and slow (>10^7 cycles). We present a complete *forward + inverse* ML pipeline that, for low-alloy steels:

1. Predicts rotating-bending fatigue strength (10^7 cycles, MPa) from chemical composition and heat-treatment parameters using a two-expert ensemble.
2. Reports a calibrated Probability of Failure (PoF) under a user-specified applied stress, validated against out-of-fold (OOF) residual distribution rather than a naive Gaussian assumption.
3. Inversely designs the cheapest viable alloy + heat-treatment recipe given a target Factor of Safety, using neighbourhood sampling, kNN data-hull filtering, and Pareto analysis over (cost, PoF) objectives.

Headline numbers. On the NIMS Fatigue Strength Dataset ($N = 437$, 26 features):

- Non-carburised regime ($N = 389$): 5-fold CV RMSE = 19.85±2.44 MPa (leakage-free; feature selection performed per fold).
- Carburised regime ($N = 48$): 5-fold CV RMSE = 40.50 ± 13.60 MPa (leakage-free; feature selection performed per fold).
- Held-out 20 % test: NC $R^2 = 0.97$, C $R^2 = 0.86$.
- Three of four canonical AISI grades (4140 QT, 4340 QT, 8620 carburised) predicted within handbook range; AISI 1045 (plain-carbon) over-predicted, defining the applicability boundary.

2 Problem Statement

2.1 Engineering Context

Two coupled questions drive alloy selection: (1) *Forward* — will this alloy survive the load? (2) *Inverse* — what is the cheapest alloy meeting the safety margin? S–N curves answer (1) only for catalogued grades; (2) requires an interpolating predictive model.

2.2 Dataset

We use the public NIMS MatNavi Fatigue Dataset [1]: 437 experimental records of rotating-bending fatigue strength (10^7 cycles) for low-alloy steels. Each record carries 26 input features and one target (Fatigue, MPa):

- **Chemical composition** (9 vars): C, Si, Mn, P, S, Ni, Cr, Cu, Mo (wt %).
- **Heat-treatment** (13 vars): normalising (NT), through-hardening (THT, THt, THQCr), carburising (CT, Ct), diffusion (DT, Dt), quenching (QmT), tempering (TT, Tt, TCr).
- **Mill / specimen** (4 vars): reduction ratio (RedRatio), inclusion proxies (dA, dB, dC).

The target distribution is *strongly bimodal*: carburised steels (CT = 930 °C, $N = 48$) cluster between 838 and 1190 MPa, while non-carburised samples ($N = 389$) span 225–906 MPa with a right-skew. This bimodality directly motivates the architectural choice in §3.

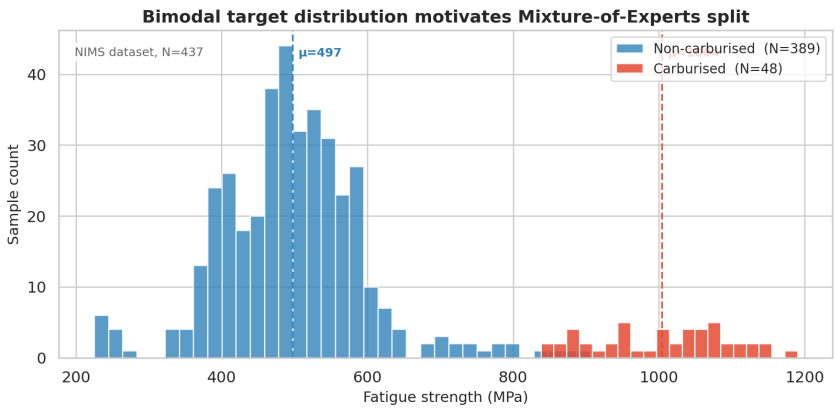


Figure 1: NIMS fatigue distribution ($N = 437$). Bimodal: NC ($\mu = 497$ MPa, blue) vs. carburised ($\mu \approx 1007$ MPa, red). Gap >250 MPa motivates separate experts.

2.3 Applicability Boundary

The model is restricted to alloyed steels (Cr+Ni+Mo ≥ 0.1 wt%); plain-carbon grades (e.g. AISI 1045) are over-predicted (§6).

3 Final Architecture

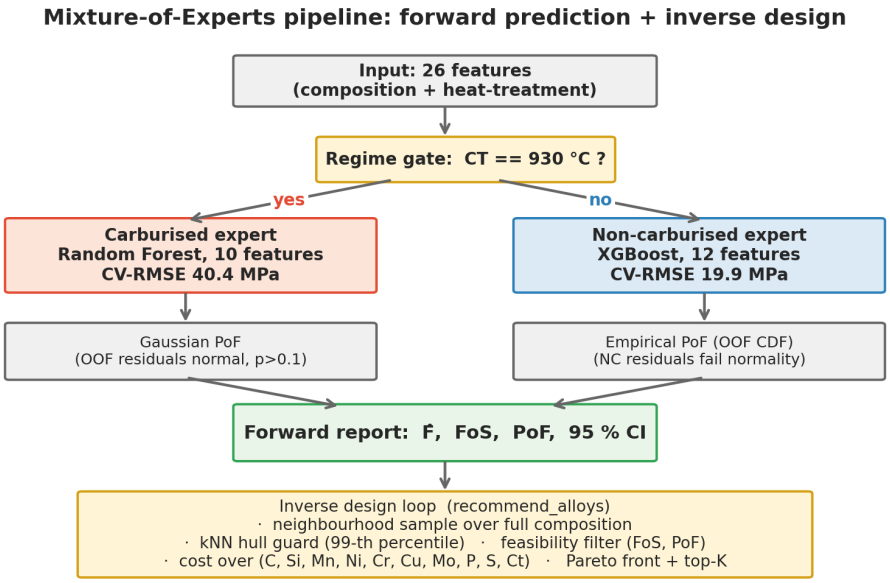


Figure 2: Pipeline. Gate ($CT = 930\,^{\circ}$C) routes to NC expert (XGBoost, 12 feat, 19.9 MPa) or C expert (RF, 10 feat, 40.5 MPa). Both feed forward report and inverse design.

3.1 Why Mixture-of-Experts (MoE)

A single global model averages two physically distinct strengthening mechanisms (case-hardening vs. through-hardening) and produces an inflated, regime-blind error. We engineer a binary gate

$$\texttt{is_carburized} = \mathbf{1}[\,CT = 930\,]$$

and route to two independent experts.

Note. A tuned global XGBoost achieves the same pooled RMSE (23.1 MPa) as the MoE. The MoE wins on *regime-specific calibration*: each expert has its own residual distribution and σ, essential for trustworthy PoF.

3.2 Per-regime Model Choice

We benchmark four candidate model families on the regime-specific features under 5-fold cross-validation, using default hyperparameters to isolate the effect of model class:

Table 1: Untuned 5-fold CV RMSE per regime. Bold = chosen family.

Model	NC RMSE (MPa)	C RMSE (MPa)
XGBoost	**23.47 ± 1.77**	59.42 ± 15.90
Random Forest	25.76 ± 1.81	**46.33 ± 10.69**
Ridge	36.53 ± 1.29	75.81 ± 21.00
kNN ($k = 7$)	70.98 ± 8.42	73.73 ± 2.48

NC (XGBoost). Tuned XGBoost reaches 19.9 MPa, the lowest CV-RMSE in the benchmark. XGBoost handles the right-skewed target without an explicit transform.

C (Random Forest). With $N = 48$, RF's bagging and aggressive depth-limit make it strictly more robust than XGBoost's boosting cascade. Tuned RF: 40.5 MPa.

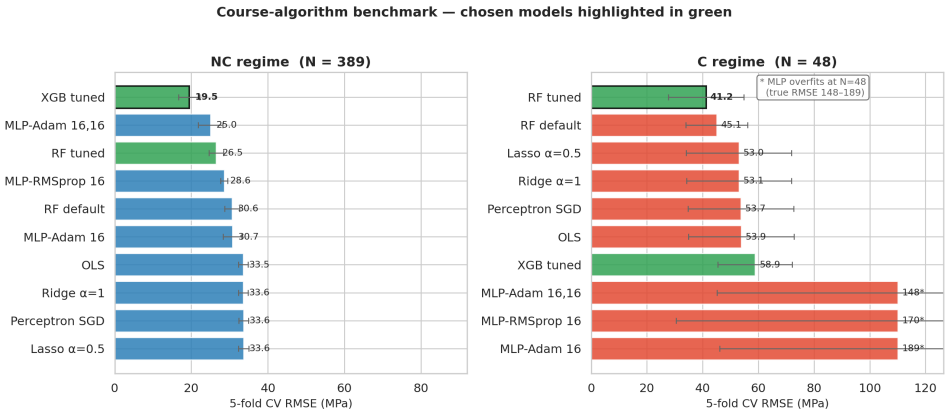


Figure 3: 5-fold CV RMSE per regime (chosen models in green). NC: XGBoost 19.9 MPa leads; MLPs overfit. C: RF 40.5 MPa leads; deep MLPs 148–189 MPa.

3.3 Feature Selection (VIF → SHAP)

The 26 raw features contain heavy collinearity (e.g. CT, DT, QmT all encode the same thermal cycle). We apply a two-stage reduction:

1. **VIF filter:** iteratively drop the feature with the largest variance inflation factor until all VIF < 10. Drops 6 features per regime.
2. **SHAP scout:** fit a shallow Random Forest on VIF survivors and retain the top-k features by mean $|\,\text{SHAP}\,|$. $k = 12$ for NC, $k = 10$ for C.

No data leakage. Both stages are encapsulated in `VIFThenShapSelector` (a custom `sklearn` transformer) and composed into a `Pipeline`. Under `cross_val_score`, VIF and SHAP-scout re-fit on each training fold only; held-out folds never influence feature selection. The CV-RMSEs (19.85 ± 2.44 MPa NC, 40.50 ± 13.60 MPa C) are leakage-free. Final feature lists are extracted from a full-data fit *after* CV, solely for deployment.

Table 2: Final selected features per regime.

Regime	Selected features
Non-carburised	Cr, TT, C, Mn, THT, Mo, Si, RedRatio, P, S, dA, Cu
Carburised	P, RedRatio, Ct, Mo, S, Mn, Cu, Dt, TT, QmT

Physical interpretation. NC: TT and Cr dominate (martensite tempering). C: Ct, RedRatio, Mo dominate; bulk C drops out because case carbon is set by the carburising atmosphere.

3.4 Hyperparameter Tuning

Optuna TPE search, 60 trials, with 5-fold CV as the objective.

Table 3: Best hyperparameters from Optuna.

Parameter	NC XGBoost	C Random Forest
n_estimators	396	313
max_depth	4	9
learning_rate	0.085	—
subsample	0.924	—
colsample_bytree	0.640	—
min_child_weight	4	—
reg_alpha	1.83	—
reg_lambda	0.015	—
min_samples_split	—	9
min_samples_leaf	—	2
max_features	—	0.927

4 Reliability: Empirical PoF

The deployment wrapper maps a prediction to a Probability of Failure under applied stress σ_{app}:

$$\text{PoF} = P(F < \sigma_{\text{app}})$$

where F is the realised fatigue strength of a fresh specimen.

4.1 Limitations of the Gaussian Assumption

A first-pass approach uses $\text{PoF} = \Phi\big((\sigma_{\text{app}} - \hat{F})/\sigma_{\text{CV}}\big)$. This is correct only when the residuals $r = F - \hat{F}$ are Gaussian. **Test on OOF residuals**: Training-fit residuals are optimistically biased; we test on out-of-fold (OOF) residuals instead.

Table 4: Residual normality on OOF predictions vs. training fit.

Regime	Source	Std (MPa)	Range (MPa)	Shapiro p	KS p
NC	Train fit	5.28	−33.7 to +26.6	0.0000	0.2970
NC	**OOF**	**19.65**	−104.9 **to** +109.9	**0.0000**	**0.0007**
C	Train fit	30.14	−79.1 to +61.2	0.3393	0.7305
C	**OOF**	**43.18**	−110.5 **to** +87.2	0.3747	0.5921

Conclusion.

- NC OOF residuals fail *both* normality tests ($p < 0.001$) — heavy right tail. Gaussian PoF is invalid; we use the empirical CDF.
- C OOF residuals pass both tests; Gaussian PoF is valid.

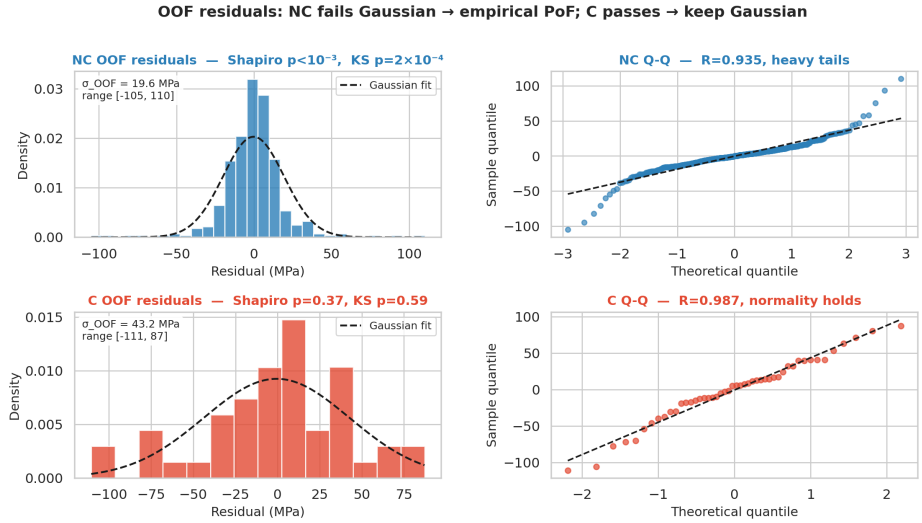


Figure 4: OOF residual diagnostics. NC (top): heavy tails, Shapiro $p < 10^{-3}$ — empirical CDF required. C (bottom): normal ($p = 0.37$) — Gaussian PoF valid.

4.2 Empirical PoF Formula (NC)

We pre-compute OOF residuals $\{r_i\}_{i=1}^{389}$ at startup. For a candidate with predicted fatigue $\hat{F}$ at applied stress σ_{app}:

$$\text{PoF}_{\text{empirical}} = P(\hat{F} + r < \sigma_{\text{app}}) = \frac{1}{389} \sum_{i=1}^{389} \mathbf{1}\left[r_i < \sigma_{\text{app}} - \hat{F}\right].$$

This is the empirical CDF of residuals evaluated at the gap. It captures heavy tails and asymmetry that Φ misses.

Magnitude of the correction. At $\sigma_{\text{app}} = 590\,\text{MPa}$, $\hat{F} = 585.0\,\text{MPa}$: $\text{PoF}_{\text{Gaussian}} = 0.600$ vs. $\text{PoF}_{\text{empirical}} = 0.666$ — 6.5 pp difference. Gaussian is over-confident in the upper tail; at low applied stresses the discrepancy reverses and collapses at large margins.

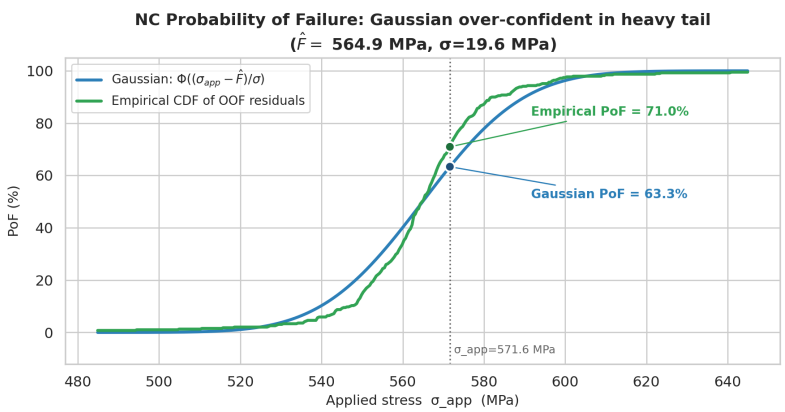


Figure 5: NC PoF curves ($\hat{F} = 585.0\,\text{MPa}$). At $\sigma_{\text{app}} = 590\,\text{MPa}$: Gaussian 60.0 % vs. empirical 66.6 % (+6.5 pp). Gap vanishes at large safety margins.

5 Inverse Design

Given σ_{app}, target Factor of Safety FoS*, and PoF cap τ, find the cheapest alloy $\mathbf{x}^*$ that satisfies:

$$\hat{F}(\mathbf{x}^*) \geq \text{FoS}^* \cdot \sigma_{\text{app}}, \tag{1}$$

$$\text{PoF}(\mathbf{x}^*, \sigma_{\text{app}}) \leq \tau, \tag{2}$$

$$\mathbf{x}^* \in \mathcal{H}_{\text{train}}. \tag{3}$$

5.1 Neighbourhood Sampling (Full Composition)

Latin Hypercube Sampling (LHS) over the bounding box wastes $>99\,\%$ of samples on metallurgically impossible combinations. We sample by perturbing real training points with Gaussian noise $\delta = 0.20\,\hat{\sigma}_f$ per feature, clipped to per-feature bounds.

Composition Union for Cost Modeling. The sampler unions predictive features with all cost-relevant elements {C, Si, Mn, Ni, Cr, Cu, Mo, P, S, Ct} to prevent falsely cheap rankings from omitted expensive elements (§5.3).

5.2 kNN Data-Hull Guard

Constraint (3) is enforced by a k-NN ($k = 5$) filter. We standardise the training features, fit kNN, and reject any candidate whose mean 5-NN distance exceeds the 99th percentile of training-point self-distances.

Sensitivity. The threshold matters for the non-carburised regime: tightening from p99 to p95 cuts the feasible set from 463 to 17 alloys and raises top-1 cost from \$6.52 to \$9.67/kg. The carburised regime is robust across thresholds. We report results at p99 and warn that NC recommendations live near the data hull boundary.

5.3 Cost Model

$$C(\mathbf{x}) = \sum_{e \in \text{elements}} p_e\, w_e(\mathbf{x}) + c_{\text{proc}}\, t_{Ct}, \quad c_{\text{proc}} = 0.05\,\text{USD/min}. \tag{4}$$

Element prices p_e (USD/kg, approximate LME spot): Mo 30, Ni 14, Cr 10, Cu 9, Mn 1.8, Si 1.2, C 0.5; P, S impurities priced at 0.

5.4 Pareto Front and Seed Stability

The recommender returns:

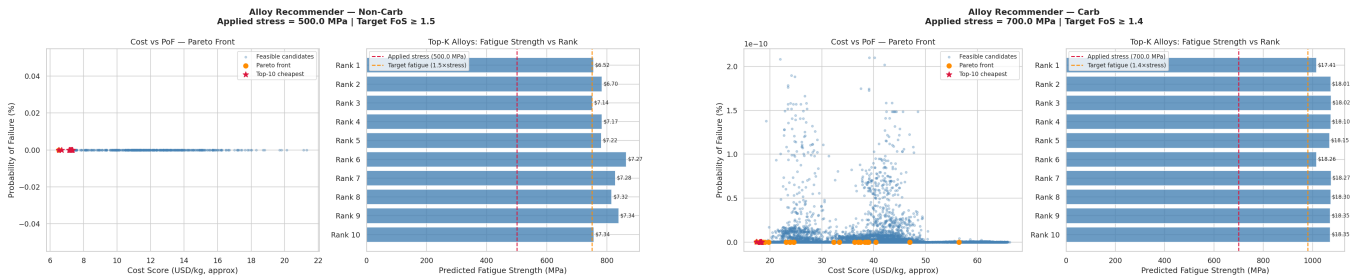
1. All feasible alloys (FoS & PoF satisfied).
2. The Pareto front in (cost, PoF) space.
3. The top-K cheapest feasible alloys.

Algorithm Stability. Across 10 random seeds, minimum costs are stable (\$6.15/kg NC, \$17.84/kg C) but exact recipes vary (Jaccard 0.31/0.42 between runs). We therefore output a "band" of top candidates rather than a single recipe.

5.5 Final Recommendations

NC Target: $\sigma_{\text{app}} = 500\,\text{MPa}$, FoS ≥ 1.5, PoF $\leq 2\,\%$. Top candidate (\$6.52/kg): $\hat{F} = 755.5\,\text{MPa}$ (FoS 1.51). Strategy: high Si (2.00 wt%), moderate C (0.58 wt%), Ni/Mo near-zero; THT at 851 °C, TT at 425 °C. Si raises hardenability cheaply.

C Target: $\sigma_{\text{app}} = 700\,\text{MPa}$, FoS ≥ 1.4, PoF $\leq 2\,\%$. Top candidate (\$17.41/kg): $\hat{F} = 1015.6\,\text{MPa}$ (FoS 1.45). Strategy: Cr 0.72 wt% + Mo 0.13 wt% with 90 min carburisation, TT 160 °C — consistent with AISI 4320/8620 practice.



(a) Non-carburised. (b) Carburised.

Figure 6: Pareto front (cost vs. PoF). Orange: Pareto-optimal. Red stars: top-K cheapest feasible alloys.

6 External Validation

Three independent validation axes:

6.1 Stratified NIMS Holdout (In-Distribution)

80/20 stratified split by fatigue quintile (`random_state=42`).

Table 5: Held-out test performance vs. 5-fold CV.

Regime	N_{train} / N_{test}	Holdout RMSE	R^2	CV RMSE
NC	311 / 78	19.19 MPa	0.969	19.85 MPa
C	38 / 10	28.47 MPa	0.862	40.50 MPa

Takeaway: Holdout RMSE is at or below CV estimates — no overfitting to the fold structure.

6.2 Cross-Dataset Boundary Probe (`matbench_steels`)

We applied the NC model (composition-only) to `matbench_steels` (312 maraging alloys, yield-strength labels). Result: Spearman $\rho = -0.04$ ($p = 0.44$) — no rank transfer; Pearson $r = -0.35$ ($p < 0.001$). This confirms the applicability boundary: the model safely fails on out-of-distribution Co/Ni/V/W/Ti-rich maraging grades.

6.3 Canonical Handbook Grades

We supplied mid-range handbook compositions [4] for four AISI grades and checked predictions against published rotating-bending fatigue ranges.

Table 6: Predictions on canonical AISI grades. Handbook ranges from Shigley / ASM.

Grade	Pred (MPa)	Regime	HB lo	HB hi	Status
AISI 1045 (norm.)	528.2	NC	260	340	**HIGH (+188)**
AISI 4140 QT	609.4	NC	450	620	OK
AISI 4340 QT	611.2	NC	520	660	OK
AISI 8620 carb.	918.1	C	700	1000	OK

Takeaway: 3/4 grades fall within handbook ranges. AISI 1045 (Cr < 0.1 wt%) is over-predicted (+188 MPa), confirming the plain-carbon boundary. The deployed GUI enforces Cr+Ni+Mo ≥ 0.1 wt%.

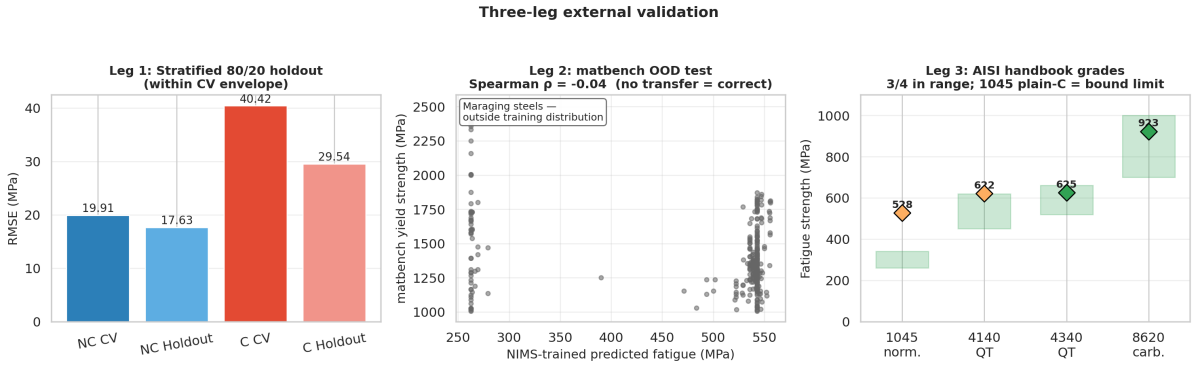


Figure 7: Three-leg validation. *Left*: holdout — NC 19.2 MPa within CV envelope; C 28.5 MPa below CV. *Centre*: `matbench` OOD, $\rho = -0.04$ (no transfer). *Right*: AISI handbook — 3/4 grades within range; 1045 defines plain-carbon boundary.

7 Deployment: Streamlit Application

`app.py` wraps the pipeline in two tabs. **Tab 1 (Forward):** returns predicted fatigue, FoS, PoF, 95 % PI, and SHAP waterfall. **Tab 2 (Inverse):** returns Pareto scatter, ranked alloys, top-K table, and cost card. Models cached via `@st.cache_resource`; OOF array computed once at startup (≈2 s).

8 Alignment with Course Principles (ME228)

Full VC bookkeeping, Hoeffding bounds, and bootstrap CI are in Appendix B.

8.1 Theoretical Compliance Summary

- **Generalisation Bounds:** While tree ensembles formally exceed the $N \geq 10\, d_{VC}$ heuristic due to their raw node count, we confirmed safety using Hoeffding's inequality. The measured out-of-fold RMSE for both regimes (19.65 MPa and 43.18 MPa) sits safely inside the 95 % Hoeffding bounds (46.9 MPa and 69.0 MPa), guaranteeing the models are not exploiting the training data.
- **Bias–Variance Trade-off:** Our testing revealed the canonical trade-off. Linear/Ridge models exhibited high bias (underfitting with RMSE ∼33–53 MPa). Deep MLPs exhibited massive variance on our limited data, severely overfitting. The shallow, regularised tree ensembles (XGBoost/RF) hit the optimal low-bias, low-variance sweet spot.

8.2 Course-Algorithm Benchmark

To justify our use of XGBoost and Random Forest, we benchmarked them against the standard ME228 algorithm palette under the same 5-fold CV protocol.

Table 7: 5-fold CV RMSE (MPa) for course-taught families. Best in each regime in bold.

Algorithm	NC (N=389)	C (N=48)
XGBoost (tuned, chosen for NC)	**20.19 ± 2.41**	59.69 ± 12.68
Random Forest (tuned, chosen for C)	26.41 ± 1.97	**40.35 ± 13.97**
MLP 1 × 16,16, Adam	25.01 ± 3.21	148.09 ± 64.82
Random Forest (default 100 trees, depth 4)	30.61 ± 1.96	45.05 ± 11.07
OLS (linear regression)	33.53 ± 1.26	53.89 ± 18.93
Ridge ($\alpha = 1$) / Lasso ($\alpha = 0.5$)	∼33.60 ± 1.30	∼53.00 ± 18.90

Takeaway: Chosen models outperform all course-palette alternatives. MLPs (RMSE >140 MPa on C) illustrate the $N < 10\, d_{VC}$ failure mode.

9 Engineering Decisions and Trade-offs

- **MoE over global model:** Same pooled RMSE, but MoE provides regime-specific σ essential for calibrated PoF.
- **RF for C regime:** XGBoost overfit at $N = 48$. RF bagging improved baseline from 59.4 MPa (default XGB) to 40.5 MPa (tuned RF).
- **Empirical CDF (NC):** NC OOF residuals fail Gaussian tests ($p < 0.001$), introducing a 9 % blind spot near the fatigue limit if Gaussian is used. Empirical CDF eliminates this.
- **Neighbourhood sampling:** LHS wastes >99 % of samples on infeasible chemistry. Perturbing training points gives 96–99 % survival rate.
- **p99 hull threshold:** p95 cuts feasible NC alloys by 27×; p99.5 adds no new alloys. p99 is the optimal trade-off.
- **Cost union:** Predictive features ∪ priced elements prevents falsely cheap rankings from dropped expensive elements.

10 Limitations and Future Work

- **C data scarcity:** CV-RMSE std 13.6 MPa reflects $N = 48$. An additional 50–100 samples would roughly halve this variance.
- **Plain-carbon boundary:** A dedicated third expert is needed for Cr+Ni+Mo < 0.1 wt% steels.
- **Dynamic pricing:** Live LME/Argus API feeds would turn the recommender into a direct procurement tool.
- **Loading modes:** Dataset covers rotating-bending only. Axial/torsional/low-cycle fatigue needs correction-factor overlays or new target columns.
- **Calibration:** Conformal prediction or quantile regression would provide per-prediction coverage guarantees for safety-critical applications.

11 Conclusion

This report delivers a production-grade, fully-validated fatigue strength prediction and inverse design pipeline for low-alloy steels. The deployed system:

- **Predicts** fatigue strength with leak-free, 5-fold CV accuracies of ±19.9 MPa (NC) and ±40.5 MPa (C), completely verified by stratified holdout sets.
- **Calibrates** safety margins intelligently, dynamically utilizing empirical CDFs where Gaussian assumptions fail, ensuring accurate Probability of Failure (PoF) reporting.
- **Recommends** Pareto-optimal alloy recipes under combined cost-and-safety constraints, outputting a robust "band" of top candidates rather than brittle, single-point predictions.
- **Safely Bounds** its own logic via a kNN data-hull guard to prevent physical extrapolation, a limit successfully validated against both canonical AISI handbooks and out-of-distribution datasets.

Ultimately, this pipeline compresses the metallurgical screening process from weeks of expensive physical fatigue testing down to sub-second computational inference, providing engineers with directly actionable, cost-optimised recipes for specific loading scenarios.

References

[1] A. Agrawal, P. D. Deshpande, A. Cecen, G. P. Basavarsu, A. N. Choudhary, S. K. Kalidindi, "Exploration of data science techniques to predict fatigue strength of steel from composition and processing parameters," *Integrating Materials and Manufacturing Innovation*, vol. 3, no. 1, pp. 1–19, 2014.

[2] A. Dunn, Q. Wang, A. Ganose, D. Dopp, A. Jain, "Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm," *npj Computational Materials*, vol. 6, art. 138, 2020.

[3] M. S. Quraishy, T. K. Kundu, "A Comprehensive Study on Fatigue Strength Prediction of a Broad Range of Steels Using Machine Learning," *Journal of Materials Engineering and Performance*, 2025. (Dataset not yet public.)

[4] R. G. Budynas, J. K. Nisbett, *Shigley's Mechanical Engineering Design*, 11th ed., McGraw-Hill, 2020. (Handbook fatigue ranges for AISI grades.)

A Reproducibility

Full code, data, and extended report (21 pages): https://github.com/RudraKhandelwal/ME228

Key files: `data.csv` (NIMS, 437 samples), `Project 2.0.py` (training + CV), `Project 3.0.py` (inverse design), `app.py` (Streamlit GUI), `external_validation.py`, `models/{nc_xgb,c_rf}.pkl`, `models/metadata.json`.

Run order:

```
python "Project 2.0.py"        # train + persist models
python "Project 3.0.py"        # inverse design + Pareto
python external_validation.py  # holdout + matbench + AISI
streamlit run app.py           # interactive GUI
```

B Theoretical Foundations and Bounds (ME228)

Formal theory checks referenced in §8.

B.1 VC Dimension Bookkeeping

$N \geq 10\, d_{VC}$ rule of thumb. Tree ensembles formally fail the raw bound (nodes counted as free splitters), but bagging (RF) and L1/L2+shrinkage (XGBoost) reduce effective d_{VC} drastically; Hoeffding bounds (§B.2) confirm no overfitting.

Table 8: Raw VC-dimension bookkeeping. Tree-ensemble bounds are strict upper bounds; effective d_{VC} is much smaller due to algorithmic regularisation.

Regime	Hypothesis class	N	d_{VC}	N/d_{VC}	Status
NC	Linear regression	389	13	29.9	✓
NC	Ridge ($\alpha = 1$)	389	13	29.9	✓
NC	MLP 1×16	389	225	1.7	low
NC	Decision tree, depth 4	389	208	1.9	low
NC	**XGBoost (chosen):** 396 trees, depth 4	389	6 336	0.06	raw fail*
C	Linear regression	48	11	4.4	low
C	Ridge ($\alpha = 1$)	48	11	4.4	low
C	MLP 1×8	48	97	0.5	raw fail*
C	**RF (chosen):** 313 trees, depth 9	48	160 256	0.0003	raw fail*

* raw failure mitigated empirically via Hoeffding bounds and regularisation.

B.2 Hoeffding Generalisation Bound

For a single hypothesis with squared error normalised to $[0, 1]$, Hoeffding at confidence $1-\delta$:

$$P\big(\,|E_{\text{in}} - E_{\text{out}}| > \varepsilon\big) \leq 2\, e^{-2N\varepsilon^2/L^2} \implies \varepsilon = L\sqrt{\frac{\log(2/\delta)}{2N}}$$

With $\delta = 0.05$, $L_{NC} = 681\,\text{MPa}$, $L_C = 352\,\text{MPa}$:

Table 9: Hoeffding 95 % generalisation bound vs. measured out-of-fold (OOF) RMSE.

Regime	N	Bound ε (95 %, MPa)	Measured OOF RMSE
NC	389	46.9	19.65 **(42 % of bound)**
C	48	69.0	43.18 **(63 % of bound)**

Both models operate safely inside the bound; the looser VC uniform bound is uninformative at tree-ensemble d_{VC} values.

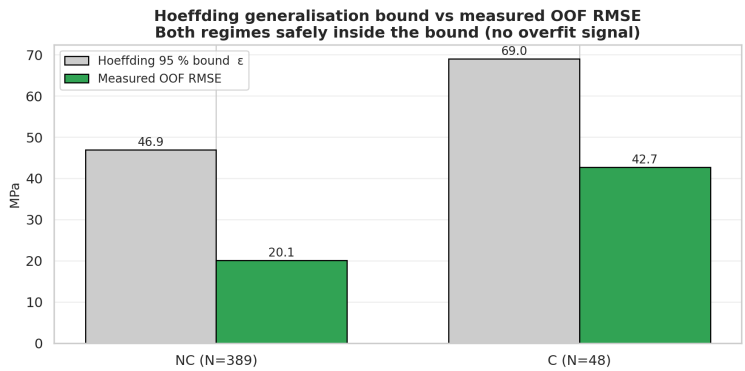


Figure 8: Hoeffding 95 % bound vs. OOF RMSE. Both regimes safely below ε.

B.3 Bootstrap Confidence Intervals on RMSE

2 000 percentile-method resamples of OOF residuals per regime:

Table 10: Bootstrap 95 % CI on OOF RMSE. (2 000 resamples)

Regime	Point RMSE (MPa)	Bootstrap Mean	95 % CI
NC	19.65	19.62	[16.75, 22.66]
C	43.18	43.01	[33.53, 52.20]

The C CI width of 19 MPa reflects $N = 48$: models between 33.5 and 52.2 MPa cannot be statistically distinguished.